SYMBIOSIS UNIVERSITY OF APPLIED SCIENCES

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School of Computer Science and Information Technology

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SYNOPSIS REPORT

ON

"AI-Enabled DoS Protection with Privacy and Post-Quantum Resilience"

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1. INTRODUCTION

1.1 Abstract of the Project

Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks are among the most disruptive forms of cyber threats, frequently targeting enterprise systems, cloud platforms, and IoT infrastructures. Traditional intrusion detection approaches often rely on static machine learning or signature-based methods, which suffer from poor scalability, lack of robustness against adversarial evasion, and vulnerability to emerging quantum computing threats.

This project introduces a **Hybrid Quantum-Resilient AI Framework** designed to ensure reliable DoS detection while incorporating **privacy-preserving mechanisms** and **post-quantum cryptography (PQC)**. The framework integrates **AI-based anomaly detection** (CNN-LSTM ensemble with adversarial training), **quantum-safe authentication protocols** (Kyber and Dilithium), and **differential privacy** for secure data handling. Additionally, a **quantum-assisted simulation** module using Qiskit demonstrates future-readiness against quantum-era challenges.

The proposed system aims to provide **high detection accuracy, robustness, scalability, and quantum resilience**, making it suitable for enterprise, ISP, and IoT environments.

1.2 Synopsis of the Project

The project develops an **intelligent intrusion detection and prevention framework** that combines deep learning, privacy preservation, and post-quantum cryptography into a unified architecture. The core of the system is a **CNN-LSTM ensemble model**, which captures both spatial and temporal traffic patterns for accurate DoS/DDoS detection. To defend against adversarial evasion, **adversarial training** and ensemble strategies are employed.

Authentication and communication security are ensured by integrating **CRYSTALS-Kyber** (for key exchange) and **Dilithium** (for digital signatures), offering resistance against both classical and quantum cryptographic attacks. To address data confidentiality, network traffic logs and training data are protected with **Differential Privacy**, ensuring compliance with privacy requirements.

For research novelty, a **quantum-inspired simulation module** based on Qiskit demonstrates how quantum feature mapping may enhance detection models in the future. This makes the framework not only practical for current deployments but also **future-proof against quantum-enabled cyber threats**.

1.3 Problem of Existing System

Current DoS detection and prevention systems face several limitations:

- **Adversarial Vulnerability:** AI/ML models can be deceived by carefully crafted malicious traffic.
- **Weak Authentication:** Reliance on RSA/ECC cryptography, which is breakable by quantum algorithms such as Shor's.
- **Lack of Privacy Preservation:** Most IDS systems store raw logs, creating privacy risks.
- **Scalability Challenges:** Existing models often fail when applied to large-scale IoT or cloud infrastructures.
- **No Quantum Preparedness:** Current solutions ignore the threat landscape of post-quantum computing.

1.4 Solution of the problem

To overcome the above challenges, the proposed project integrates multiple defense layers:

- **AI-based robust detection:** A CNN-LSTM ensemble with adversarial training ensures high detection accuracy and resilience against adversarial attacks.
- **Post-Quantum Cryptography (PQC):** Kyber for key exchange and Dilithium for authentication safeguard against future quantum attacks.
- **Privacy-preserving mechanisms:** Differential Privacy protects sensitive data in logs and training datasets.
- **Quantum-assisted readiness:** A Qiskit-based simulation module showcases how quantum feature maps can improve detection performance in the future.

2. PROJECT MONITORING SYSTEM INTRODUCTION

2.1 Functional Requirement

- **FR1: Real-time DoS/DDoS Detection** – Detect and classify malicious traffic patterns using CNN-LSTM ensemble models.

- **FR2: Adversarial Robustness** – Employ adversarial training and ensemble learning to defend against evasion attacks.

- **FR3: Quantum-Safe Authentication** – Integrate Kyber (Key Encapsulation Mechanism) and Dilithium (Digital Signatures) for secure communication resistant to quantum attacks.

- **FR4: Privacy-Preserving Data Handling** – Ensure log storage and model training with Differential Privacy to protect sensitive information.

- **FR5: Quantum Simulation Capability** – Use Qiskit to simulate quantum feature mapping for future-readiness and research extension.

- **FR6: Traffic Control Mechanism** – Provide automated decision-making to block malicious traffic and allow legitimate requests.

2.2 Non-Functional Requirement

- **Performance:** Ensure low-latency detection suitable for real-time environments.

- **Scalability:** Support high-volume network traffic across cloud and IoT ecosystems.

- **Security:** Provide resilience against adversarial traffic and ensure quantum-safe authentication.

- **Reliability:** Maintain high detection accuracy with minimal false positives/negatives.

- **Privacy:** Preserve user and organizational data confidentiality with differential privacy techniques.

- **Usability:** System should be user-friendly for network administrators and easily deployable in real-world environments.

- **Maintainability:** Modular design for updates, upgrades, and integration of future quantum-resistant algorithms.

2.3 Process Model

The proposed process model follows a **layered workflow** to ensure end-to-end protection:

1. **Traffic Capture:** Collect real-time traffic using network monitoring tools or benchmark datasets (e.g., CICDDoS2019, UNSW-NB15).

2. **Preprocessing & Feature Extraction:** Clean, normalize, and extract flow-level features.

3. **AI Detection (CNN-LSTM Ensemble):** Analyze patterns to classify traffic as benign or malicious.

4. **Adversarial Defense Layer:** Apply adversarial training and ensemble strategies for robustness.

5. **PQC Authentication:** Use Kyber for key exchange and Dilithium for digital signatures to authenticate communication.

6. **Privacy-Preserving Logging:** Apply Differential Privacy to logs for secure data storage.

7. **Quantum Simulation Module:** Simulate quantum feature maps using Qiskit for future enhancements.

8. **Decision Engine:** Final decision → **Allow legitimate traffic / Block malicious traffic**.

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3. SOFTWARE REQUIREMENT SPECIFICATION

3.1 Software Requirement

To implement the proposed Hybrid Quantum-Resilient AI Framework, the following software components are required:

Programming Languages & Environment

- **Python 3.10+ :** Core programming language for model development and integration.
- **Jupyter Notebook / PyCharm :** IDEs for experimentation, testing, and development.

Libraries & Frameworks

- **Deep Learning & ML:** PyTorch / TensorFlow (for CNN-LSTM and adversarial training).
- **Classical ML & Preprocessing:** Scikit-learn (metrics, feature selection, preprocessing).
- **Privacy Mechanisms:** Opacus / diffprivlib (for Differential Privacy).
- **Post-Quantum Cryptography (PQC):** pqcrypto library implementing Kyber & Dilithium.
- **Quantum Simulation:** Qiskit (for quantum-assisted simulation and feature mapping).
- **Explainable AI (XAI):** SHAP, Captum (to interpret model decisions).

Database & Storage

- **MySQL / SQLite :** Storage for traffic logs and system records.

Operating System

- **Windows 11 (for development/testing).**

Version Control & Collaboration

- **GitHub :** For source code management and collaborative development.

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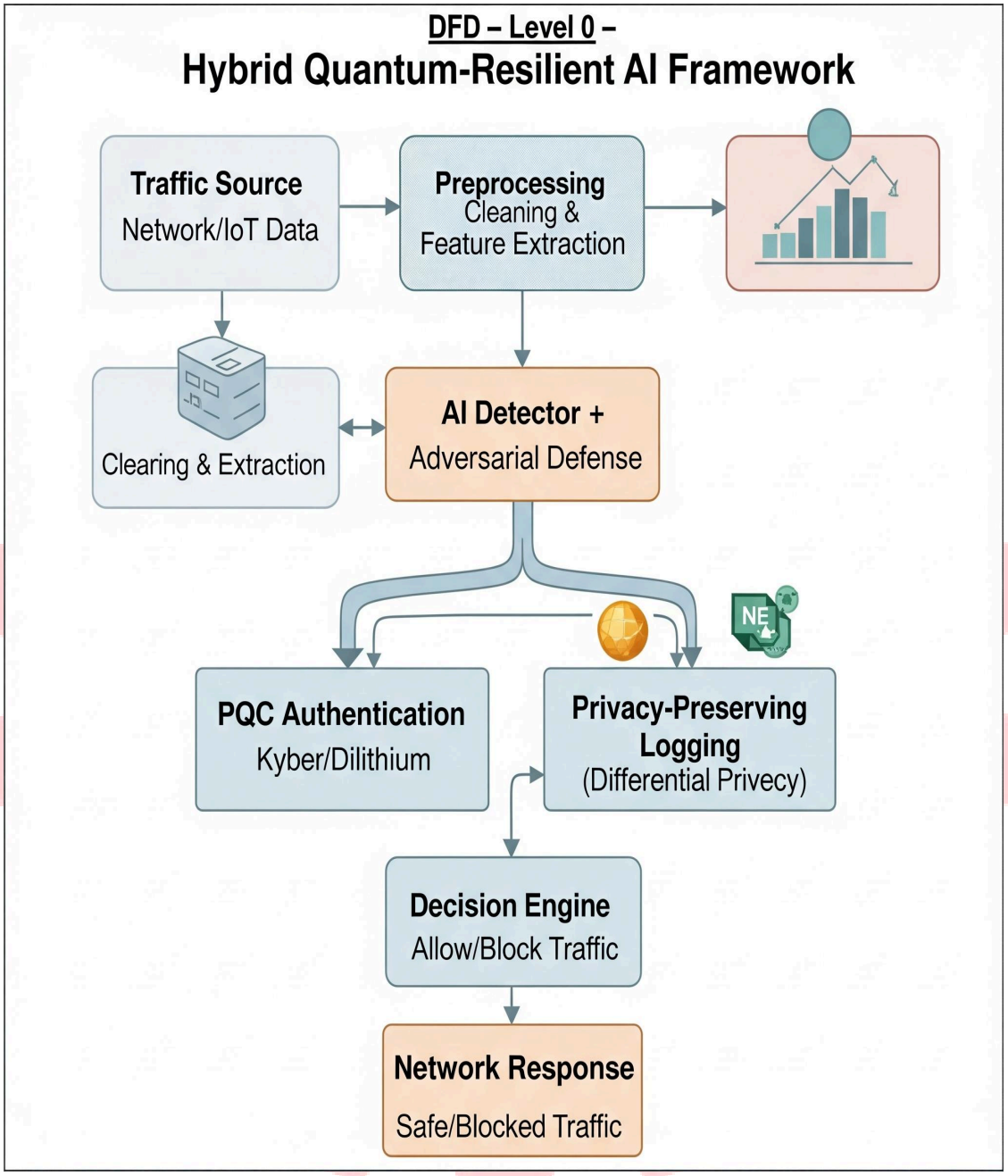
4. SYSTEM ANALYSIS

4.1 Data Flow Diagram:



5. CONCLUSION

The proposed project, **Hybrid Quantum-Resilient AI Framework for DoS Detection**, addresses the pressing challenges of modern cybersecurity by combining **robust AI detection, post-quantum cryptography, and privacy-preserving mechanisms** into a unified defense architecture. Unlike existing systems that remain vulnerable to adversarial evasion, weak cryptographic foundations, and privacy breaches, this solution introduces **multi-layered resilience** against both current and future threats.

The integration of a **CNN-LSTM ensemble model with adversarial training** enhances detection accuracy and robustness against sophisticated attacks. The adoption of **post-quantum cryptographic algorithms (Kyber and Dilithium)** ensures that authentication and key management remain secure even in the era of quantum computing. Furthermore, the inclusion of **differential privacy-based logging** provides confidentiality for sensitive network data, addressing a key limitation of present-day intrusion detection systems.

An additional novelty is the **quantum-assisted simulation module**, which demonstrates how quantum feature mapping can be leveraged in the future to further improve anomaly detection. This forward-looking design makes the framework **future-proof, scalable, and adaptable** for enterprise networks, Internet Service Providers (ISPs), and IoT ecosystems.

In conclusion, the project delivers a **comprehensive, scalable, and quantum-resilient DoS detection framework** that not only mitigates today's threats but also anticipates the security challenges of the post-quantum era.